Zagazig University
Faculty of Computers & Informatics
Mid Term Exam Academic Year: 2024-2025

Department of Computer Science
Study Year/Level: First Year
Course Title: Discrete Structures
Course Code: CS103

Model 1

Examination Date: 06/04/2025
Examination Starts: Morning
Allowed Examination Time: 1 Hour
Total Marks: 15

Student Name: **Student ID:**

- **Authorized Materials:**
- **Calculators allowed. No other additional materials or aides are allowed.**
- **Instructions to Invigilators:**
- **The exam paper may be taken out of the examination room after halftime.**
- **Instructions to Students:**
- **If you feel ill, let us know immediately.**
- **Answer All of the following questions.**
- **If you get stuck on a question, move on, and come back later if you have time.**
- **Remember to time yourself.**

Enjoy!

1. Determine the cardinality of each of the following sets. **(2 points)**

a) $\{x \mid x \in \mathbf{Z} \text{ and } x^2 < 10\}$

b) $\{x \mid x \in \mathbf{N} \text{ and } 9x^2 - 1 = 0\}$

c) $P(\{1, 2, 3, 4, 5\})$

d) $A \times B$, where $A = \{a, b, c\}$ and $B = \varnothing$

2. In the questions below suppose A = {1, 2, 3, 4, 5}. Mark the statement TRUE or FALSE. **(2 points)**

a) $\{1\} \in P(A)$

b) $\{\varnothing\} \subseteq P(A)$

c) $\{2, 4\} \in A \times A$

d) $(1, 1) \in A \times A$

3. Write an **explicit** formula for each of the following sequences: **(2 points)**

Answer:

a) 1, 1/3, 1/5, 1/7, 1/9, …

b) 2, 0, 2, 0, 2, 0, 2, ….

4. Suppose R and S are relations on {a, b, c, d}, where R = {(a, b), (a, d), (b, c), (c, c), (d, a)} and S = {(a, c), (b, d), (d, a)}. Construct R^2, S^3, $S \circ R$, S^∞. **(3 Points)**

Answer:

a) $R^2 =$

b) $S^3 =$

c) $S \circ R =$

d) $S^\infty =$

5. Let A = {1, 2, 3, 4}. If $M_R = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ (**3 Points**)

a) Draw the digraph for the relation defined by the matrix M_R.

Answer:

b) Is R irreflexive? (True or False)

c) Is R symmetric? (True or False)

d) Is R antisymmetric? (True or False)

e) Is R asymmetric? (True or False)

f) Is R transitive? (True or False)

6. Let A = {1, 2, 3, 4} and let R = {(2, 3), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)}. Find the transitive closure of the relation R using Warshall algorithm. (**3 Points**)

Answer:

Good Luck *Dr. Abduallah Gamal*